

NAVYA SIVA RAJ

Software Engineer, AI Engineer | Python, Node.js, Vue.js, AWS

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EDUCATION

Master of Science, Computer Science · GPA: 3.8/4.0

Aug 2024 – May 2026

University of Houston, Houston, TX

Relevant Coursework: Cloud Computing, Software Design, Database Management, Machine Learning

Bachelor of Technology, Computer Science and Engineering (Honors) · GPA: 9.57/10.0

Nov 2020 – Jun 2024

Rajiv Gandhi Institute of Technology (RIT), Kottayam, Kerala, India

TECHNICAL SKILLS

- **Programming:** Python, Java, C/C++, JavaScript
- **Backend & APIs:** Playwright, FastAPI, Flask, Node.js, Django, REST APIs, JSON, Web Services
- **Databases:** MySQL, MongoDB
- **DevOps & Cloud:** Git, Docker, CI/CD, Linux, AWS, Jelastic (Reclaim Cloud)
- **Web Development:** React.js, Vue.js, HTML, CSS, Bootstrap
- **Artificial Intelligence & Machine Learning:** AutoGen, LangGraph, Scikit-learn, TensorFlow, PyTorch, NLP, Large Language Models (BERT, GPT), Prompt Engineering, LoRA

WORK EXPERIENCE

Application Developer | Hewlett Packard Enterprise Data Science Institute at UH

Jun 2025 – Present

- Automated semester report generation using **Node.js**, **Vue.js**, and **MongoDB**, reducing creation time 30–44 minutes per report and eliminating manual workflows.
- Led design and initial development of a consent-based public sharing feature enabling 1,100+ students to share profiles and projects, boosting their professional visibility.
- Strengthened application reliability by implementing bi-weekly MongoDB backups and a global error-handling framework, ensuring data recovery within a 2-week window and improving system observability.
- Configured and deployed full-stack application with NGINX on **Jelastic (Reclaim Cloud)**, supporting 1,100+ active users in a production environment.

Research Assistant | University of Houston, Sugarland

Oct 2024 – Jul 2025

- Built NLP preprocessing pipelines to clean and standardize 10,000+ social media posts, improving data quality and consistency.
- Performed large-scale data acquisition from social media forums using **Python**, constructing a structured dataset from 6+ months of forum data.
- Applied sentiment analysis and topic modeling to surface emerging trends, delivering insights to research faculty.

Machine Learning Intern | Indian Institute of Information Technology, Kottayam (IIITK)

Dec 2021 – Apr 2022

- Collaborated with 6 interns under the guidance of IIITK's Data Research Group to engineer and deploy machine learning models using **Python**, **NumPy**, **Pandas**, and **Matplotlib**, covering supervised, unsupervised, and reinforcement learning techniques.
- Produced data visualizations and optimized data pipelines that improved model performance and interpretability.

ACADEMIC PROJECT EXPERIENCE

Agentic Moodle LMS Automation System | University of Houston

Jan 2026 – Present

- Designed an agentic browser automation system for Moodle LMS that autonomously logs in, identifies pending assignments, and generates/submits responses.
- Evolved the system from rule-based scripts to an LLM-driven, fully autonomous multi-agent architecture (**AutoGen**) with tool-based browser actions (**Playwright**) and HTML-to-action decision making using GPT-4o/Qwen.
- Orchestrated a scalable **FastAPI** backend with asynchronous automation and **Docker**-based Moodle integration, enabling end-to-end workflow execution.

BioVault - Biometric Authentication System | RIT, Kottayam

Sep 2023 – Apr 2024

- Engineered a secure biometric authentication system with a team of 4, distributing encrypted biometric templates across multiple servers to ensure fault-tolerant data protection.
- Developed backend services and REST APIs using **Flask** and **MongoDB**, handling processing and retrieval of biometric data.
- Integrated image processing pipelines (**OpenCV**) for fingerprint feature extraction.

My Simple TezFunds - Crowdfunding Platform | TezAsia'22 Hackathon

Jul 2022 – Aug 2022

- Architected and launched a web-based crowdfunding platform for women-led businesses within a 3-member team.
- Deployed smart contracts using **SmartPy** for secure contribution tracking and fund transfers.
- Delivered dynamic frontend components using **React.js** with integrated payment workflows.

Multi-Class Multi-Label Traffic Crash Cause Identification | University of Houston

Jan 2025 – May 2025

- Constructed an end-to-end ML pipeline on 794K+ Chicago crash records, consolidating 40 contributory causes into 5 target classes and reducing 48 raw features to 18 key predictors via feature selection.
- Evaluated multiple classifiers (e.g., Decision Tree, Logistic Regression, kNN) for a multi-label classification task using standard metrics (accuracy, precision, recall, and F1-score).
- Optimized kNN hyperparameters to achieve 54.19% accuracy, a 15-point gain over the 39% baseline, making it the top-performing model across all evaluated classifiers.